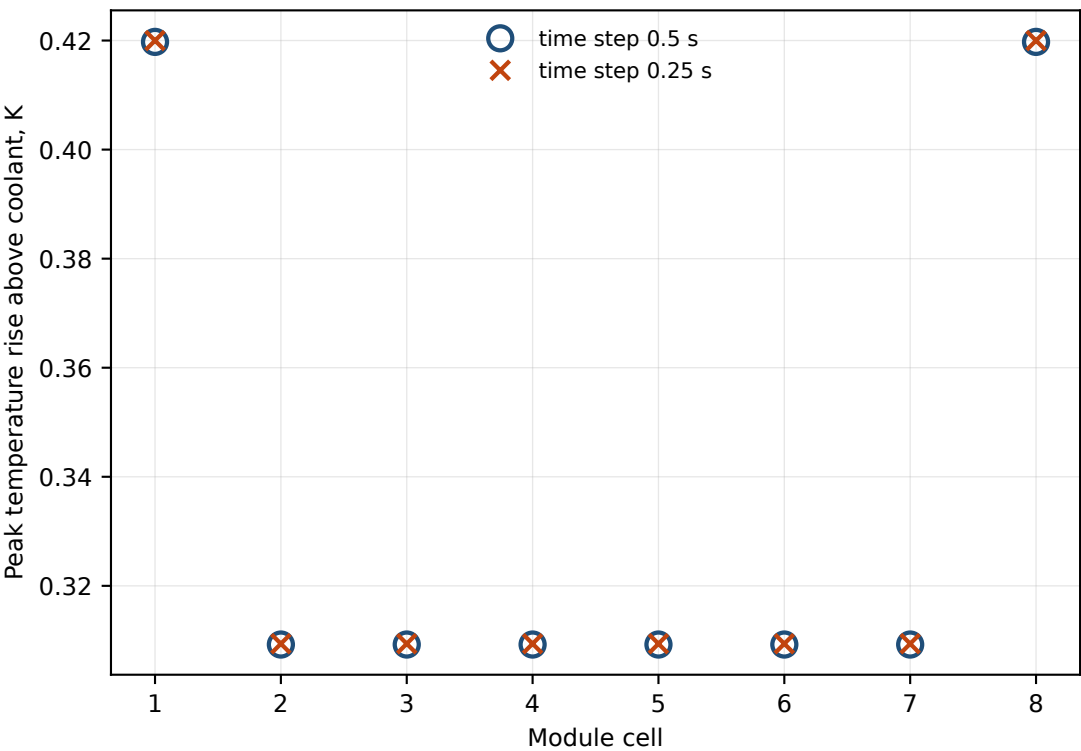


Act C, C6.4 — does the answer depend on the time step? The same mesh, the same duty cycle, two step sizes.

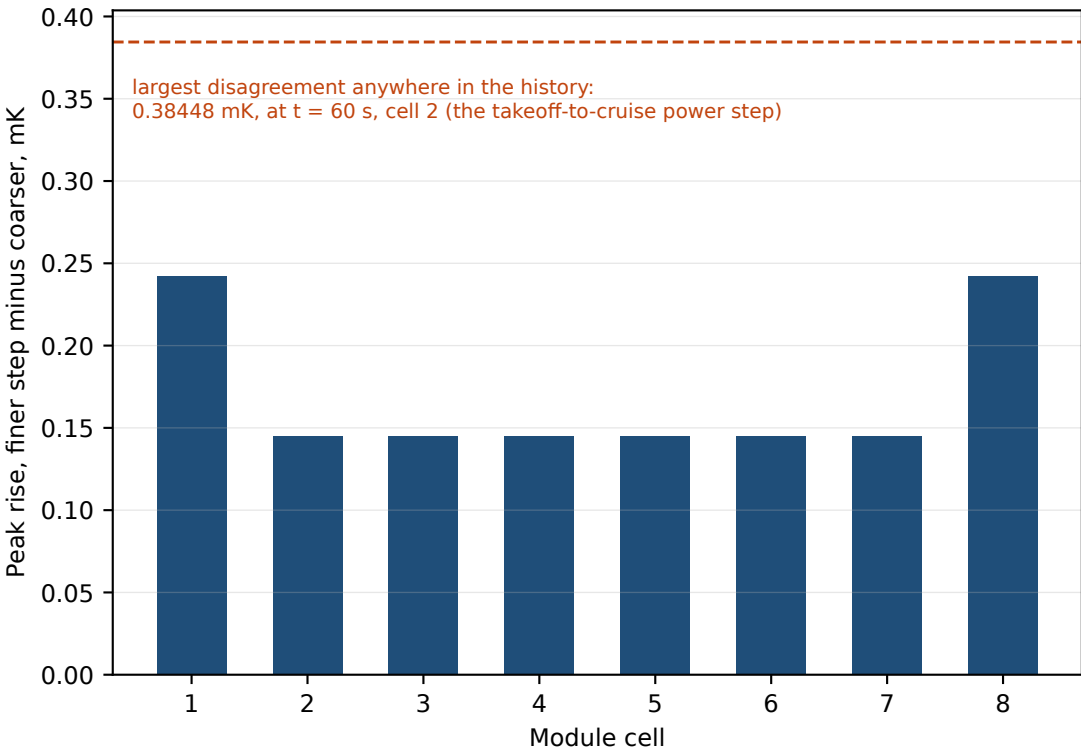
The two step sizes, drawn on one axis.

The markers sit on top of each other at this scale: that overlay is the result.



The same eight numbers, subtracted.

Note the axis: the whole range of this panel is under three ten-thousandths of a kelvin.



Quantity	Time step 0.5 s	Time step 0.25 s	Change	Change, %
Peak rise, end cells 1 and 8, K	0.419751	0.419993	+2.4223e-04	+0.0577
Peak rise, interior cells 2-7, K	0.309239	0.309384	+1.4516e-04	+0.0469
Largest disagreement at any of the 181 write times, K	—	—	3.8448e-04	at t = 60 s, cell 2
Heat input left unaccounted by the energy budget, %	0.145859	0.073031	—	—

Halving the solver time step from 0.5 s to 0.25 s changed the module's 15-minute peak temperature rise by under 0.0003 K on every one of the eight cells — 0.058 % on the two end cells and 0.047 % on the six interior cells — and the largest disagreement at any point in the 900-second history was 0.00038 K, at the takeoff-to-cruise power step. Both step sizes close the energy budget to better than 99.85 % of the heat put in, the finer step leaving 0.073 % unaccounted against the coarser step's 0.146 %. This is a two-point step-size check on an ungated feasibility run: it shows the answer barely moves when the step is halved, and it is not a convergence study, carries no error bar and grades nothing.